

Dispersal and metapopulations

Learning objectives:

1. Understand source-sink dynamics
2. Model occurrence in a patchy landscape
3. Use dispersal kernels to model dispersal limitation

Environmental heterogeneity, including spatial, can affect population growth rate

Connect multiple populations in a heterogeneous environment

Patch dynamics & metapopulations

- Caused by environmental heterogeneity
- Good patches/bad patches -> source-sink dynamics

Draw lattice with patches

Source-sink dynamics:

- Source patches: $F > m$ *excess of seeds*
- Sink patches: $F < m$

Circle plants in source patches

these are what we call metapopulations

- metapopulations: population of populations

question: how do the metapopulations in sink patches persist? what's required so they don't go extinct?

Dispersal

- local dispersal *global dispersal* - seeds can get anywhere
- max. dispersal distance, closer = more seeds
- dispersal kernels - 3D distributions

Draw circle around individual plant in lattice

Plot distance from parent with bell curve

- dispersal limitation can hinder population growth in patchy environments

Competition-colonization tradeoff

- reproductive tradeoff in patchy landscapes
- tradeoff between competitive ability and dispersal distance
- number and traits of seeds

draw lattice, competitive species can survive in worse patches, colonizing species gets to next good patch

Modeling

if going incidence function route:

- Assume seeds can get anywhere in landscape
- Incidence: fraction of patches occupied by a plant species = J
- Extinction events on occupied patches
- Colonization of unoccupied patches
- $dJ/dt = \text{gains colonization} - \text{losses extinction}$
- $C = \text{colonization rate}$
- $E = \text{extinction rate}$
- $1-J$ is fraction of empty patches
- $J = \text{source patches}$

$$\frac{dJ}{dt} = c(1 - J)J - eJ$$

- Equilibrium patch occupancy is when colonization rate equals extinction rate